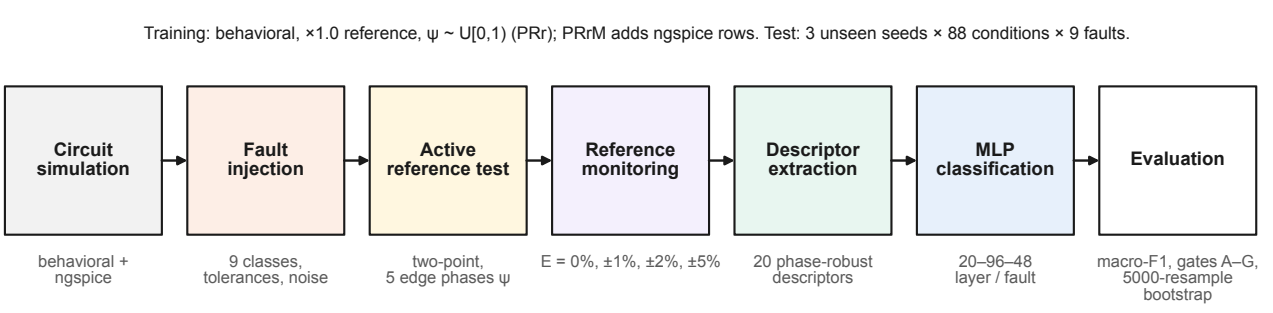
Training: behavioral, ×1.0 reference, ψ ~ U[0,1) (PRr); PRrM adds ngspice rows. Test: 3 unseen seeds × 88 conditions × 9 faults.
Circuit simulation
Fault injection
Active reference test
Reference monitoring
Descriptor extraction
MLP classification
Evaluation
behavioral + ngspice
9 classes, tolerances, noise
two-point, 5 edge phases ψ
E = 0%, ±1%, ±2%, ±5%
20 phase-robust descriptors
20–96–48 layer / fault
macro-F1, gates A–G, 5000-resample bootstrap